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Analysis of the Input Stages of a USB Charger

Mains Supply, Step-Down Transformer and Bridge Rectifier

Notation

Symbol	Meaning	Value used
V_p	Mains (primary) voltage, RMS	220 V
$V_{p,pk}$	Mains peak voltage	311.1 V
V_s	Transformer secondary voltage, RMS (measured)	10.53 V
$V_{s,pk}$	Secondary peak voltage	14.89 V
V_D	Forward voltage drop of one diode	0.7 V
$V_{o,pk}$	Rectifier output peak voltage	13.49 V
$V_{o,avg}$	Rectifier output average (DC) voltage	8.59 V
$V_{o,rms}$	Rectifier output RMS voltage	9.54 V
f	Mains frequency	50 Hz
f_o	Rectifier output (ripple) frequency	100 Hz
N_p, N_s	Number of turns, primary and secondary	–

Subscripts: p = primary, s = secondary, pk = peak, rms = root-mean-square, avg = average, o = output.

1. Mains Input

The mains supply is a sinusoidal voltage of RMS value $V_p = 220$ V and frequency f = 50 Hz. Its peak value follows from the relation between peak and RMS values of a sinusoid:

$$V_{p,pk} = V_p \times \sqrt{2} = 220 \times 1.4142 \approx 311.1 \text{ V} \quad (1)$$

2. Step-Down Transformer

For an ideal transformer, the ratio of the voltages equals the ratio of the number of turns:

$$V_s / V_p = N_s / N_p \quad (2)$$

With the measured secondary voltage $V_s = 10.53$ V, the effective turns ratio is:

$$N_p / N_s = V_p / V_s = 220 / 10.53 \approx 20.9 : 1 \quad (3)$$

The peak value of the secondary voltage is:

$$V_{s,pk} = V_s \times \sqrt{2} = 10.53 \times 1.4142 \approx 14.89 \text{ V} \quad (4)$$

The transformer changes only the amplitude; the frequency remains 50 Hz. It also provides galvanic isolation, meaning there is no direct electrical connection between the mains and the output circuit.

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3. Bridge Rectifier

The bridge rectifier consists of four diodes (D1 to D4). During the positive half-cycle D1 and D2 conduct, and during the negative half-cycle D3 and D4 conduct. In both cases current flows through the load in the same direction, so the output is a full-wave rectified (|sin|) waveform.

At any instant two diodes are in series with the load, so the output peak is reduced by two diode drops (V_D = 0.7 V for silicon diodes such as the 1N4007 or 1N5408):

$$V_{o,pk} = V_{s,pk} - 2V_D = 14.89 - 2(0.7) = 13.49 \text{ V} \quad (5)$$

The average (DC) value of a full-wave rectified sinusoid is:

$$V_{o,avg} = 2V_{o,pk} / \pi = 2 \times 13.49 / 3.1416 \approx 8.59 \text{ V} \quad (6)$$

The RMS value of the rectified output is:

$$V_{o,rms} = V_{o,pk} / \sqrt{2} = 13.49 / 1.4142 \approx 9.54 \text{ V} \quad (7)$$

Since each half-cycle of the input becomes a positive pulse, the output frequency is double the mains frequency:

$$f_o = 2f = 2 \times 50 = 100 \text{ Hz} \quad (8)$$

The output therefore repeats every 1/100 = 10 ms.

Diode rating. Each reverse-biased diode must withstand the peak inverse voltage (PIV), which is approximately $V_{s,pk}$ = 14.89 V. This is far below the 1000 V rating of the 1N4007, so the diodes are safe. For higher load currents, the 1N5408 (3 A) is preferred over the 1N4007 (1 A).

4. Summary of Results

Quantity	Symbol	Value
Mains peak voltage	$V_{p,pk}$	311.1 V
Secondary voltage (RMS)	V_s	10.53 V
Secondary voltage (peak)	$V_{s,pk}$	14.89 V
Turns ratio	$N_p : N_s$	20.9 : 1
Rectifier output (peak)	$V_{o,pk}$	13.49 V
Rectifier output (average)	$V_{o,avg}$	8.59 V
Rectifier output (RMS)	$V_{o,rms}$	9.54 V
Output frequency	f_o	100 Hz

The rectifier output is still pulsating DC, falling to zero every 10 ms. A smoothing capacitor is therefore required in the next stage to reduce the ripple.